

Bergman Kernels at Singular Points and Hilbert–Samuel Multiplicity

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Abstract

Let X be a normal projective variety of pure dimension n , let L be an ample line bundle, and let h be a smooth strictly positive Hermitian metric on L in the sense of complex spaces. Writing $\omega = c_1(L, h)$ and using the intrinsic volume form $\omega^n/n!$ on X_{reg} , we study the ordinary Bergman function of $H^0(X, L^m)$. Each Bergman function is smooth on X in the natural ambient complex-space sense. We prove uniform two-sided estimates

$$cm^n \leq B_m(x) \leq Cm^n$$

for every $x \in X$ and all sufficiently large m . If

$$C_m^{\text{opt}} := \max_{x \in X} \frac{B_m(x)}{m^n}$$

denotes the best global upper constant, then $C_m^{\text{opt}} = 1 + O(m^{-1})$. More precisely, we obtain the uniform one-sided estimate

$$\frac{B_m(x)}{m^n} \leq \frac{1 + O(m^{-1})}{\text{mult}_x X},$$

and at every fixed point $x \in X$ we prove

$$\lim_{m \rightarrow \infty} m^{-n} B_m(x) = \frac{1}{\text{mult}_x X}.$$

Thus the coefficient at a singular point is the reciprocal, not the value, of the Hilbert–Samuel multiplicity. The proof combines a sharp Lelong–Jensen mean-value inequality with ambient coherent states and the Gaussian scaling of the local analytic density. We also explain why this coarse-smooth coefficient is generally unrelated to the normalized volume of a klt singularity, and why changing the measure or passing to an orbifold metric leads to a different answer.

1 Introduction

On a smooth polarized manifold (X, L, h) , the Tian–Catlin–Zelditch expansion gives, under the normalization used below,

$$B_m(x) = m^n + O(m^{n-1}).$$

For a singular projective variety, one still expects the usual local behavior away from the singular locus, but no such expansion can extend uniformly across the singular set in general. The purpose of this note is to identify the leading pointwise coefficient at a fixed singular point when both the Hermitian weight and the volume form are smooth in the coarse complex-space sense.

Although X is singular, the individual functions B_m have no regularity defect at the singular set in this coarse sense. Indeed, each B_m is locally the restriction of a smooth ambient function; in particular, it is continuous on X and smooth in the usual sense on X_{reg} . This elementary observation is recorded in Proposition 2.2. It should not be confused with uniform regularity as $m \rightarrow \infty$: the pointwise limit of $m^{-n}B_m$ is generally discontinuous across the singular locus.

There are two points that should be emphasized at the outset. First, the correct coefficient is the reciprocal of the Hilbert–Samuel multiplicity. For example, at the vertex of the projective quadric cone the multiplicity is 2, while the normalized Bergman function converges to $1/2$. Second, the choice of measure is essential. The conclusion below concerns the matched curvature volume $\omega^n/n!$. It need not hold for an adapted canonical measure, nor for a nondegenerate orbifold measure on a local uniformizing cover.

The proof has three ingredients.

- (i) A Lelong–Jensen inequality applied to the positive psh current $|f|^2[X]$ gives a lower bound for the norm of every holomorphic section with prescribed value. The Lelong number of this current is exactly $\text{mult}_x X$. After Gaussian rescaling this yields

$$\|s\|_m^2 \geq (\text{mult}_x X - o(1))m^{-n}|s(x)|_{h^m}^2.$$

A uniform ambient Bochner normalization sharpens the error to $\text{mult}_x X(1 - O(m^{-1}))$, uniformly in x .

- (ii) After replacing L by a fixed very ample power, the given metric can be extended, with its trace preserved exactly, to a smooth positive metric on $\mathcal{O}_{\mathbb{P}^N}(1)$. A normalized coherent state in the ambient projective space restricts to a global section on X and provides the reverse estimate.
- (iii) Thie’s density theorem identifies the radial mass at x with the degree of the tangent cone, namely $\text{mult}_x X$. Its Gaussian rescaling identifies the sharp coefficient.

The Lelong–Jensen argument also gives a sharp description of the global upper envelope. A uniform ambient Bochner normal form yields

$$\frac{B_m(x)}{m^n} \leq \frac{1 + O(m^{-1})}{\text{mult}_x X}$$

simultaneously for all $x \in X$. Since regular points have multiplicity one and the Hilbert polynomial controls the average of B_m , the optimal global upper constant satisfies $C_m^{\text{opt}} = 1 + O(m^{-1})$. Thus the sharp asymptotic upper constant is 1, even though $m^{-n}B_m$ itself does not converge uniformly across the singular locus.

The klt assumption is not used in the analytic proof except insofar as it guarantees normality and supplies the setting in which normalized volume is defined. We state the result for a normal projective variety because that is the natural situation of interest.

2 Conventions and statements of the main results

Throughout the note,

$$dd^c = \frac{\sqrt{-1}}{2\pi} \partial\bar{\partial}.$$

If the metric is initially written as $h_0 e^{-u}$, we simply denote this resulting metric by h ; the hypotheses and all Bergman norms below concern the combined metric, not the particular choice of h_0 and u . With this convention, if $q(z) = \pi \sum_{j=1}^N |z_j|^2$ on $\mathbb{C}^N$, then

$$\beta := dd^c q = \frac{\sqrt{-1}}{2} \sum_{j=1}^N dz_j \wedge d\bar{z}_j,$$

and

$$\int_{\mathbb{C}^n} e^{-\pi|z|^2} \frac{\beta^n}{n!} = 1.$$

Definition 2.1 (Smooth strict positivity on a complex space). *Let X be a reduced complex space and let L be a line bundle on X . A Hermitian metric h on L is called smooth and strictly positive if, for every $x \in X$, there are a neighborhood U of x , a closed holomorphic embedding $j : U \hookrightarrow \Omega \subset \mathbb{C}^N$, a holomorphic frame e of L on U , and a function $\varphi \in C^\infty(\Omega)$ such that*

$$|e|_h^2 = e^{-\varphi}|_U, \quad dd^c \varphi > 0 \quad \text{on } \Omega.$$

The restrictions of the local forms $dd^c \varphi$ define a smooth Kähler form on X , denoted by $c_1(L, h)$.

This is stronger than requiring positivity merely on X_{reg} and is a substantive hypothesis near a singular point.

Let X be a normal projective variety of pure dimension n , let L be an ample line bundle, and let h be a smooth strictly positive metric on L . Set

$$\omega = c_1(L, h), \quad dV_\omega = \frac{\omega^n}{n!} \quad \text{on } X_{\text{reg}}.$$

For $m \geq 1$, equip

$$\mathcal{H}_m := H^0(X, L^m)$$

with the norm

$$\|s\|_m^2 = \int_{X_{\text{reg}}} |s|_{h^m}^2 dV_\omega.$$

The singular set has zero dV_ω -measure, so this is a genuine finite-dimensional Hilbert norm. If $\{s_{m,j}\}_{j=1}^{N_m}$ is an orthonormal basis, define

$$B_m(x) = \sum_{j=1}^{N_m} |s_{m,j}(x)|_{h^m}^2.$$

Proposition 2.2 (Smoothness of the Bergman function). *For every $m \geq 1$, the function B_m is smooth on X in the ambient complex-space sense. More precisely, near every point of X it is the restriction of a smooth function on a local ambient manifold. No global generation assumption on L^m is needed.*

Proof. Fix $x \in X$ and use Definition 2.1 to choose a local embedding $j : U \hookrightarrow \Omega \subset \mathbb{C}^N$, a holomorphic frame e of L on U , and a smooth ambient weight φ satisfying $|e|_h^2 = e^{-\varphi}|_U$. Write

$$s_{m,j}|_U = f_{m,j} e^{\otimes m}, \quad f_{m,j} \in \mathcal{O}_X(U).$$

After shrinking U , the local quotient description $\mathcal{O}_U \simeq \mathcal{O}_\Omega / \mathcal{I}_U$ allows us to lift every $f_{m,j}$ to a holomorphic function $F_{m,j}$ on Ω . Hence

$$B_m|_U = \left(e^{-m\varphi} \sum_{j=1}^{N_m} |F_{m,j}|^2 \right) \Big|_X.$$

The expression in parentheses is smooth on Ω , proving the claim. The formula is independent of the chosen frame and orthonormal basis. $\square$

Equivalently,

$$B_m(x) = \sup_{0 \neq s \in H^0(X, L^m)} \frac{|s(x)|_{h^m}^2}{\|s\|_m^2}. \quad (1)$$

For all sufficiently large m , ampleness makes L^m globally generated. We may then put

$$\lambda_m(x) := B_m(x)^{-1} = \inf \{\|s\|_m^2 : |s(x)|_{h^m} = 1\}. \quad (2)$$

Write

$$e_x(X) := e(\mathcal{O}_{X,x}, \mathfrak{m}_x) = \text{mult}_x X$$

for the Hilbert–Samuel multiplicity of the maximal ideal.

Theorem 2.3 (Uniform estimates and the optimal upper coefficient). *There exist constants $c, C > 0$ and $m_0 \geq 1$, depending only on (X, L, h) , such that*

$$cm^n \leq B_m(x) \leq Cm^n$$

for every $x \in X$ and every $m \geq m_0$. Moreover, if

$$C_m^{\text{opt}} := \max_{x \in X} \frac{B_m(x)}{m^n}, \quad (3)$$

then

$$C_m^{\text{opt}} = 1 + O(m^{-1}). \quad (4)$$

More precisely, there is a constant $K > 0$, depending only on (X, L, h) , such that

$$\frac{B_m(x)}{m^n} \leq \frac{1 + K/m}{e_x(X)} \quad (5)$$

for every $x \in X$ and every $m \geq m_0$.

Theorem 2.4 (Pointwise singular asymptotic). *For every fixed $x \in X$,*

$$\lim_{m \rightarrow \infty} \frac{B_m(x)}{m^n} = \frac{1}{e_x(X)}. \quad (6)$$

Equivalently,

$$\lim_{m \rightarrow \infty} m^n \lambda_m(x) = e_x(X).$$

At a smooth point $e_x(X) = 1$, so the leading coefficient agrees with the usual Tian–Catlin–Zelditch coefficient.

Remark 2.5 (Normalization of the measure). *If the Hilbert norm is instead defined using $d\mu = a dV_\omega$, where a is continuous and strictly positive at x , the same local argument gives*

$$\lim_{m \rightarrow \infty} m^{-n} B_{m,\mu}(x) = \frac{1}{a(x)e_x(X)}.$$

In particular, if $d\mu = dV_\omega/V$, where $V = \int_X dV_\omega$, the limit is $V/e_x(X)$.

Remark 2.6 (No uniform convergence across the singular locus). *Each function $m^{-n} B_m$ is smooth in the sense of Proposition 2.2, whereas $x \mapsto e_x(X)^{-1}$ equals 1 on X_{reg} and is generally strictly smaller at a singular point. Consequently, the convergence in Theorem 2.4 cannot be uniform on a set containing both regular points and a singular point of multiplicity greater than one.*

3 Exact extension of the metric to projective space

The construction of sharp global peak sections requires an ambient metric whose restriction is exactly the prescribed metric. A merely bounded comparison is not enough: after taking the m -th power it would introduce an exponential error.

Lemma 3.1 (A partition with controlled derivatives). *Let K be a compact subset of a complex manifold M , and let $\{U_i\}_{i=1}^r$ be a finite open cover of a neighborhood of K . After shrinking that neighborhood, there is a smooth partition of unity $\{\chi_i\}$ subordinate to $\{U_i\}$ such that*

$$\chi_i \geq 0, \quad \sum_i \chi_i = 1, \quad |\partial \chi_i|_\beta^2 \leq C \chi_i$$

for a fixed background Hermitian form β and a uniform constant C .

Proof. Choose smooth nonnegative functions ρ_i with $\text{supp } \rho_i \subseteq U_i$ such that $\sum_i \rho_i^2 > 0$ near K , and put

$$\chi_i = \frac{\rho_i^2}{\sum_j \rho_j^2}.$$

On a sufficiently small fixed neighborhood of K , the denominator is bounded below. Differentiating the quotient shows $|\partial \chi_i|^2 \leq C \rho_i^2$, hence $|\partial \chi_i|^2 \leq C' \chi_i$. At a zero of χ_i , its first derivative also vanishes, so the inequality extends continuously. $\square$

Proposition 3.2 (Exact positive ambient extension). *Choose $r \geq 1$ such that L^r is very ample, and let*

$$\iota : X \hookrightarrow \mathbb{P}^N, \quad \iota^* \mathcal{O}_{\mathbb{P}^N}(1) \simeq L^r$$

be the associated embedding. *There exists a smooth strictly positive Hermitian metric H on $\mathcal{O}_{\mathbb{P}^N}(1)$ such that*

$$H|_X = h^r.$$

Consequently, if $\Omega = c_1(\mathcal{O}(1), H)$, then

$$\Omega|_{X_{\text{reg}}} = r\omega.$$

Proof. Fix a smooth positive reference metric H_0 on $\mathcal{O}(1)$ and set $\theta = c_1(\mathcal{O}(1), H_0)$. Write

$$h^r = H_0|_X e^{-\varphi},$$

where $\varphi \in C^\infty(X)$ in the sense of complex spaces. By Definition 2.1, the required positive local extensions can be taken in the fixed ambient space $\mathbb{P}^N$. Here is the verification. Near $p \in X$, choose a local embedding $j : (X, p) \hookrightarrow (\mathbb{C}^M, 0)$ on which the full local weight of h^r extends to a smooth strictly psh function ψ . The coordinate functions of j extend holomorphically to a neighborhood of p in $\mathbb{P}^N$; denote the resulting map by F . If $g_1, \dots, g_s$ generate the ideal of X in that neighborhood, then

$$(dF, dg_1, \dots, dg_s) : T_p \mathbb{P}^N \longrightarrow \mathbb{C}^M \oplus \mathbb{C}^s$$

is injective: the common kernel of the dg_α is the Zariski tangent space $T_p X$, and $dF|_{T_p X} = dj$ is injective. Consequently,

$$\psi \circ F + A \sum_{\alpha=1}^s |g_\alpha|^2$$

is strictly psh at p for $A > 0$ sufficiently large, and it has the same restriction to X as the full weight of h^r . Subtracting the local weight of H_0 produces a relative potential extending

φ . Shrinking and using compactness, we thus obtain finitely many open sets $U_i \subset \mathbb{P}^N$ meeting X and functions $\varphi_i \in C^\infty(U_i)$ such that

$$\varphi_i|_{X \cap U_i} = \varphi, \quad \theta + dd^c \varphi_i \geq \delta \beta$$

after shrinking the sets, for a background Hermitian form β and some $\delta > 0$ independent of i . Choose the partition from Lemma 3.1 and, for $\eta > 0$, define near X

$$\Phi_\eta = \eta \log \left(\sum_i \chi_i e^{\varphi_i/\eta} \right). \quad (7)$$

Since all active φ_i have the same restriction to X ,

$$\Phi_\eta|_X = \varphi.$$

We verify that strict positivity is preserved. Fix $p \in X$ and an ambient $(1, 0)$ vector ξ . Put

$$a_i = \partial \varphi_i(\xi), \quad \bar{a} = \sum_i \chi_i a_i, \quad b_i = \partial \chi_i(\xi).$$

A direct differentiation of (7), using $\sum_i \partial \chi_i = \sum_i \partial \bar{\partial} \chi_i = 0$, gives at p

$$\begin{aligned} (\theta + dd^c \Phi_\eta)(\xi, \bar{\xi}) &= \sum_i \chi_i (\theta + dd^c \varphi_i)(\xi, \bar{\xi}) + \frac{c_0}{\eta} \sum_i \chi_i |a_i - \bar{a}|^2 \\ &\quad + 2c_0 \operatorname{Re} \sum_i b_i \overline{(a_i - \bar{a})}, \end{aligned} \quad (8)$$

where $c_0 > 0$ is the harmless constant determined by the convention for dd^c . Weighted Cauchy–Schwarz and $|\partial \chi_i|^2 \leq C \chi_i$ imply

$$2 \left| \sum_i b_i \overline{(a_i - \bar{a})} \right| \leq \frac{1}{2\eta} \sum_i \chi_i |a_i - \bar{a}|^2 + C_1 \eta |\xi|_\beta^2.$$

It follows that

$$\theta + dd^c \Phi_\eta \geq (\delta - C_2 \eta) \beta \quad \text{on } X.$$

For one fixed sufficiently small η , compactness and continuity give a neighborhood W of X on which

$$\theta + dd^c \Phi_\eta > 0. \quad (9)$$

It remains to globalize this function without changing it near X . If $X = \mathbb{P}^N$, the preceding neighborhood is already the whole ambient compact manifold after taking the finite cover, so there is nothing further to do. Assume henceforth that $X \subsetneq \mathbb{P}^N$. For $d \gg 1$, the coherent sheaf $\mathcal{I}_X(d)$ is globally generated. Choose generators $F_1, \dots, F_M$; their common zero scheme has support exactly X . On $\mathbb{P}^N \setminus X$, put

$$\rho = \frac{1}{d} \log \left(\sum_{j=1}^M |F_j|_{H_0^d}^2 \right).$$

Then ρ is smooth on $\mathbb{P}^N \setminus X$, tends to $-\infty$ precisely along X , and

$$\theta + dd^c \rho \geq 0.$$

For $0 < \varepsilon < 1$, set $\rho_\varepsilon = (1 - \varepsilon)\rho$. Thus

$$\theta + dd^c \rho_\varepsilon = \varepsilon \theta + (1 - \varepsilon)(\theta + dd^c \rho) > 0. \quad (10)$$

Choose nested neighborhoods $X \subset W_1 \Subset W_2 \Subset W$. By adding a constant to ρ_ε , arrange that $\rho_\varepsilon + C > \Phi_\eta + 2\tau$ on an outer collar of ∂W_2 . After this constant has been fixed, shrink W_1 so that $\rho_\varepsilon + C < \Phi_\eta - 2\tau$ on W_1 . On $W_2 \setminus X$, take a convex regularized maximum

$$M_\tau(\Phi_\eta, \rho_\varepsilon + C)$$

which agrees exactly with the larger branch whenever the two arguments differ by at least τ . It therefore glues smoothly to Φ_η on W_1 and to $\rho_\varepsilon + C$ outside W_2 . Translation invariance of M_τ and its convexity show that the resulting global function Φ satisfies $\theta + dd^c\Phi > 0$. Since $\Phi = \Phi_\eta$ near X , one also has $\Phi|_X = \varphi$. The metric

$$H = H_0 e^{-\Phi}$$

has all the required properties. $\square$

4 A sharp local mean-value inequality

We next prove an asymptotically sharp uniform upper bound for the Bergman function, as well as one half of the pointwise asymptotic.

Lemma 4.1 (Normal form of the weight). *Fix $x \in X$. In a local embedding centered at x , after changing the holomorphic frame of L and making a complex linear change of ambient coordinates, the local weight of h can be written as*

$$\varphi(z) = q(z) + O(|z|^3), \quad q(z) = \pi|z|^2. \quad (11)$$

Moreover, if $\beta = dd^c q$, then as ambient forms

$$\omega = \beta + O(|z|). \quad (12)$$

Proof. Subtract from the weight the real part of a holomorphic polynomial of degree at most two. This removes the constant, linear, and pluriharmonic quadratic terms without changing the curvature. Since the ambient Levi form is positive definite at x , a complex linear coordinate change identifies its Hermitian quadratic part with $\pi|z|^2$. Taylor's theorem then gives (11), and differentiation gives (12). $\square$

Lemma 4.2 (Uniform ambient Bochner normal form). *There exist constants $A, T > 0$, depending only on (X, L, h) , with $AT \leq 1/2$, such that the following holds. For every $x \in X$, one can choose a local ambient embedding centered at x , ambient holomorphic coordinates z , and a holomorphic frame e_x of L for which*

$$|e_x|_h^2 = e^{-\varphi_x}, \quad \varphi_x(0) = 0, \quad q(z) = \pi|z|^2, \quad \beta = dd^c q,$$

and, on $\{q < T\}$,

$$\varphi_x \leq q + Aq^2, \quad dd^c \varphi_x \geq (1 - Aq)\beta \quad (13)$$

as an inequality of ambient Hermitian forms.

Proof. Choose finitely many relatively compact local ambient embeddings, frames of L , and strictly psh ambient extensions of their weights as in Definition 2.1. Their Levi forms have uniform positive lower and upper bounds, and their derivatives through order four are uniformly bounded.

At a moving center x , first make a complex linear change of coordinates that identifies the Hermitian quadratic part of the weight with $\pi|z|^2$. Changing the frame removes the

pluriharmonic Taylor terms through degree three. A holomorphic quadratic coordinate change then removes the remaining terms of bidegree $(2, 1)$ and $(1, 2)$. This is the standard Bochner normalization and gives

$$\varphi_x(z) = q(z) + R_x(z), \quad |R_x(z)| \leq C|z|^4, \quad |dd^c R_x(z)|_\beta \leq C|z|^2.$$

The estimates for the second derivatives follow directly from Taylor's theorem and the uniform fourth-derivative bounds; they are not obtained by formally differentiating the first estimate. The coefficients of all the coordinate and frame changes are uniformly bounded. The finite cover and the quantitative inverse function theorem therefore give a common coordinate radius. After enlarging the constant and shrinking this radius, the displayed estimates imply (13) with constants A, T independent of x . $\square$

Lemma 4.3 (Lelong–Jensen mean-value inequality). *Let $(X, 0) \subset (\mathbb{C}^N, 0)$ be a reduced pure n -dimensional analytic germ, let $q = \pi|z|^2$, and let $\beta = dd^c q$. If $f \in \mathcal{O}_{X,0}$, then, for all sufficiently small $t > 0$,*

$$\int_{X_{\text{reg}} \cap \{q < t\}} |f|^2 \frac{\beta^n}{n!} \geq \text{mult}_0 X |f(0)|^2 \frac{t^n}{n!}. \quad (14)$$

Consequently, for $T > 0$ in that range and $m \geq 1$,

$$\begin{aligned} & \int_{X_{\text{reg}} \cap \{q < T\}} |f|^2 e^{-mq} \frac{\beta^n}{n!} \\ & \geq \text{mult}_0 X |f(0)|^2 m^{-n} \left(1 - e^{-mT} \sum_{j=0}^{n-1} \frac{(mT)^j}{j!} \right). \end{aligned} \quad (15)$$

Proof. Extend f to an ambient holomorphic function $\tilde{f}$ and consider the current

$$T = |\tilde{f}|^2 [X].$$

It is a positive plurisubharmonic current of bidimension (n, n) , because

$$dd^c T = dd^c |\tilde{f}|^2 \wedge [X] \geq 0.$$

The Lelong–Jensen formula for positive psh currents implies that its projective mass

$$t^{-n} \int_{X_{\text{reg}} \cap \{q < t\}} |f|^2 \frac{\beta^n}{n!}$$

is nondecreasing in t ; see Demailly [4]. Thie's density theorem [8], together with $|f(z)|^2 = |f(0)|^2 + o(1)$ on X , identifies its limit at the origin as

$$\frac{\text{mult}_0 X |f(0)|^2}{n!}.$$

This proves (14). Notice that the argument is uniform in f : no derivative bound for f enters the inequality, so f may depend on an external parameter such as m .

Let

$$F(t) = \int_{X_{\text{reg}} \cap \{q < t\}} |f|^2 \frac{\beta^n}{n!}.$$

Stieltjes integration by parts and (14) give

$$\begin{aligned} \int_{\{q < T\}} |f|^2 e^{-mq} \frac{\beta^n}{n!} &= \int_0^T e^{-mt} dF(t) \\ &= e^{-mT} F(T) + m \int_0^T e^{-mt} F(t) dt \\ &\geq \text{mult}_0 X |f(0)|^2 \int_0^T e^{-mt} d\left(\frac{t^n}{n!}\right), \end{aligned}$$

which is exactly (15). $\square$

Proposition 4.4 (Sharp lower bound for the extremal norm). *For every fixed $x \in X$,*

$$\liminf_{m \rightarrow \infty} m^n \lambda_m(x) \geq e_x(X). \quad (16)$$

Equivalently,

$$\limsup_{m \rightarrow \infty} m^{-n} B_m(x) \leq e_x(X)^{-1}.$$

Proof. Use the coordinates and frame in Lemma 4.1. If $s = f e_L^m$ and $|s(x)|_{h^m} = 1$, the normalization can be chosen so that $f(0) = 1$. Put

$$T_m = \frac{(\log m)^2}{m}.$$

On $\{q < T_m\}$ one has

$$m|\varphi - q| = O\left(\frac{(\log m)^3}{\sqrt{m}}\right) = o(1)$$

and, by (12),

$$\frac{\omega^n}{n!} = (1 + o(1)) \frac{\beta^n}{n!}$$

uniformly on the regular part of this set. Lemma 4.3 therefore gives

$$\begin{aligned} \|s\|_m^2 &\geq \int_{X_{\text{reg}} \cap \{q < T_m\}} |f|^2 e^{-mq} \frac{\omega^n}{n!} \\ &\geq (1 - o(1)) \int_{X_{\text{reg}} \cap \{q < T_m\}} |f|^2 e^{-mq} \frac{\beta^n}{n!} \\ &\geq (e_x(X) - o(1)) m^{-n}. \end{aligned}$$

The error is independent of the particular section s , so taking the infimum proves the proposition. $\square$

The preceding pointwise estimate admits the following uniform quantitative refinement.

Proposition 4.5 (Uniform asymptotically sharp upper estimate). *There are constants $K > 0$ and $m_1 \geq 1$, depending only on (X, L, h) , such that*

$$\frac{B_m(x)}{m^n} \leq \frac{1 + K/m}{e_x(X)} \quad (17)$$

for all $x \in X$ and $m \geq m_1$. Moreover, the optimal upper constants from (3) satisfy

$$C_m^{\text{opt}} = 1 + O(m^{-1}).$$

Proof. Fix $x \in X$ and use the coordinates and frame supplied by Lemma 4.2. Write a global section locally as $s = fe_x^{\otimes m}$. Since $\varphi_x(0) = 0$,

$$|f(0)|^2 = |s(x)|_{h^m}^2.$$

Set

$$F(t) = \int_{X_{\text{reg}} \cap \{q < t\}} |f|^2 \frac{\beta^n}{n!}.$$

Lemma 4.3 gives, for $0 < t < T$,

$$F(t) \geq e_x(X) |s(x)|_{h^m}^2 \frac{t^n}{n!}. \quad (18)$$

The range of t is uniform in x , because the q -balls remain inside the fixed finite collection of ambient charts.

Restricting the ambient form inequality in (13) to $T_z X_{\text{reg}}$ gives

$$\frac{\omega^n}{n!} \geq (1 - Aq)^n \frac{\beta^n}{n!}.$$

Consequently,

$$\begin{aligned} \|s\|_m^2 &\geq \int_{X_{\text{reg}} \cap \{q < T\}} |f|^2 e^{-m\varphi_x} \frac{\omega^n}{n!} \\ &\geq \int_{X_{\text{reg}} \cap \{q < T\}} |f|^2 e^{-mq - Amq^2} (1 - Aq)^n \frac{\beta^n}{n!}. \end{aligned} \quad (19)$$

Let

$$g_m(t) = e^{-mt - Amt^2} (1 - At)^n, \quad 0 \leq t \leq T.$$

Since $AT \leq 1/2$, this function is positive and strictly decreasing. If

$$F_0(t) = e_x(X) |s(x)|_{h^m}^2 \frac{t^n}{n!},$$

then (18), Stieltjes integration by parts, and $g'_m \leq 0$ imply

$$\begin{aligned} \int_0^T g_m dF &= g_m(T)F(T) - \int_0^T F(t)g'_m(t) dt \\ &\geq g_m(T)F_0(T) - \int_0^T F_0(t)g'_m(t) dt = \int_0^T g_m dF_0. \end{aligned}$$

It follows from (19) that

$$\|s\|_m^2 \geq e_x(X) |s(x)|_{h^m}^2 m^{-n} J_m, \quad (20)$$

where

$$J_m = \frac{1}{(n-1)!} \int_0^{mT} e^{-u - Au^2/m} \left(1 - \frac{Au}{m}\right)^n u^{n-1} du.$$

We claim that $J_m \geq 1 - C_0/m$ for a constant independent of x and m . Indeed, put

$$\rho_n(u) = \frac{e^{-u} u^{n-1}}{(n-1)!}, \quad R_m = \sqrt{\frac{m}{4A}}.$$

For all sufficiently large m , one has $R_m < mT$ and $Au/m \leq 1$ on $[0, R_m]$. On this interval,

$$e^{-Au^2/m} \left(1 - \frac{Au}{m}\right)^n \geq 1 - \frac{Au^2}{m} - \frac{nAu}{m}.$$

Using

$$\int_0^\infty u \rho_n(u) du = n, \quad \int_0^\infty u^2 \rho_n(u) du = n(n+1),$$

and the exponentially small tail $\int_{R_m}^\infty \rho_n(u) du = O(e^{-c\sqrt{m}})$, we obtain

$$J_m \geq 1 - \frac{C_0}{m}. \quad (21)$$

Combining (20) and (21), and then applying (1), proves

$$\frac{B_m(x)}{m^n} \leq \frac{1}{e_x(X)(1 - C_0/m)} \leq \frac{1 + 2C_0/m}{e_x(X)}$$

for $m \geq 2C_0$. This proves (17).

It remains to identify the optimal global constant. Since $e_x(X) \geq 1$, the estimate just proved gives

$$C_m^{\text{opt}} \leq 1 + \frac{K}{m}.$$

On the other hand, if

$$V = \int_X dV_\omega = \frac{L^n}{n!},$$

then orthonormality and the Hilbert polynomial yield

$$\int_X B_m dV_\omega = h^0(X, L^m) = Vm^n + O(m^{n-1}).$$

Therefore

$$C_m^{\text{opt}} \geq \frac{h^0(X, L^m)}{Vm^n} \geq 1 - \frac{K'}{m}.$$

Together with the upper bound, this proves $C_m^{\text{opt}} = 1 + O(m^{-1})$. $\square$

5 Ambient coherent states and the uniform lower estimate

Let $M = \mathbb{P}^N$, let $A = \mathcal{O}_M(1)$, and let H be the metric furnished by Proposition 3.2. Write $\Omega = c_1(A, H)$ and equip $H^0(M, A^k)$ with the L^2 norm defined by $\Omega^N/N!$. Let $P_k(y, x)$ denote the Bergman projection kernel and $B_k^M(x)$ its diagonal Bergman function.

Lemma 5.1 (Normalized ambient coherent states). *For every $x \in M$ and every $k \gg 1$, there exists $S_{k,x} \in H^0(M, A^k)$ such that*

$$|S_{k,x}(x)|_{H^k} = 1$$

and the following estimates hold uniformly in x :

(a) for all $y \in M$,

$$|S_{k,x}(y)|_{H^k}^2 \leq Ce^{-c\sqrt{k}d_\Omega(x,y)};$$

(b) in Kähler normal coordinates and a preferred frame at x ,

$$|S_{k,x}(x + w/\sqrt{k})|_{H^k}^2 \longrightarrow e^{-\pi|w|^2}$$

locally uniformly for bounded w .

Proof. For a unit vector $v \in A_x^k$, set

$$S_{k,x}(y) = \frac{P_k(y, x)v}{B_k^M(x)}.$$

Since $P_k(x, x) = B_k^M(x) \text{Id}_{A_x^k}$, the value at x is v . The diagonal expansion $B_k^M(x) = k^N + O(k^{N-1})$, the smooth near-diagonal scaling expansion, and the global off-diagonal estimate

$$|P_k(y, x)| \leq Ck^N e^{-c\sqrt{k}d_\Omega(x,y)}$$

give (a) and (b). These estimates are standard for a smooth positive line bundle on a compact Kähler manifold; see [3, Theorem 3.18'] for the scaling expansion and [7, Theorem 0.1] for the global exponential estimate. $\square$

Lemma 5.2 (Uniform volume growth of the analytic cycle). *There are constants $C_0, r_0 > 0$ such that*

$$\int_{X_{\text{reg}} \cap B_\Omega(x,t)} \frac{(\Omega|_X)^n}{n!} \leq C_0 t^{2n}$$

for every $x \in X$ and $0 < t < r_0$.

Proof. In a fixed ambient coordinate chart, the integration current $[X]$ is a positive closed current and the Euclidean mass ratio is controlled by the Lelong monotonicity formula [6]. For radii below a fixed chart radius, the mass ratio at a smaller radius is bounded above by the ratio at the fixed outer radius. Smooth metric comparison changes the estimate by a fixed factor. A finite covering of the compact set X gives uniform constants. $\square$

Proposition 5.3 (Uniform global peak sections). *There are constants $C > 0$ and m_2 such that, for every $m \geq m_2$ and every $x \in X$, one can find $s_{m,x} \in H^0(X, L^m)$ satisfying*

$$|s_{m,x}(x)|_{h^m} = 1, \quad \|s_{m,x}\|_m^2 \leq Cm^{-n}.$$

Consequently,

$$B_m(x) \geq C^{-1}m^n.$$

Proof. First let $m = rk$. Restrict $S_{k,x}$ from Lemma 5.1 to X . Since $H|_X = h^r$ and $\Omega|_X = r\omega$, Lemmas 5.1 and 5.2, followed by integration over annuli of width $k^{-1/2}$, give

$$\begin{aligned} \|S_{k,x}|_X\|_{rk}^2 &= r^{-n} \int_{X_{\text{reg}}} |S_{k,x}|_{H^k}^2 \frac{(\Omega|_X)^n}{n!} \\ &\leq Ck^{-n} \leq C'(rk)^{-n}. \end{aligned}$$

The constants are uniform in x .

It remains to treat all residue classes modulo r . For each $j \in \{0, \dots, r-1\}$ choose an integer $a_j \equiv j \pmod{r}$ sufficiently large that L^{a_j} is globally generated. A fixed finite generating set and compactness give, for every $x \in X$, a section $\tau_{j,x} \in H^0(X, L^{a_j})$ such that

$$|\tau_{j,x}(x)|_{h^{a_j}} = 1, \quad \sup_X |\tau_{j,x}|_{h^{a_j}} \leq C_j,$$

where C_j is independent of x . If $m = a_j + rk$, take

$$s_{m,x} = \tau_{j,x}(S_{k,x}|_X).$$

Its value at x has norm one and

$$\|s_{m,x}\|_m^2 \leq C_j^2 Ck^{-n} \leq C'm^{-n}.$$

The extremal formula (1) finishes the proof. $\square$

Propositions 4.5 and 5.3 prove Theorem 2.3.

6 Local analytic density and the exact pointwise coefficient

We now sharpen Proposition 5.3 at one fixed point. The only information about the tangent cone that is needed is its radial density, which is exactly the Hilbert–Samuel multiplicity.

Lemma 6.1 (Radial Gaussian scaling). *Let $(X, 0) \subset (\mathbb{C}^N, 0)$ be a reduced pure n -dimensional analytic germ, put $q = \pi|z|^2$ and $\beta = dd^c q$, and write $e = \text{mult}_0 X$. Then*

$$F(t) := \int_{X_{\text{reg}} \cap \{q < t\}} \frac{\beta^n}{n!} = e \frac{t^n}{n!} + o(t^n) \quad (t \downarrow 0). \quad (22)$$

Consequently, for every fixed $A > 0$,

$$\lim_{k \rightarrow \infty} k^n \int_{X_{\text{reg}} \cap \{q < A/k\}} e^{-kq} \frac{\beta^n}{n!} = e \left(1 - e^{-A} \sum_{j=0}^{n-1} \frac{A^j}{j!} \right). \quad (23)$$

Proof. Thie’s theorem identifies the density of an analytic cycle with its geometric multiplicity [8]; this is the degree of the tangent-cone cycle and equals the Hilbert–Samuel multiplicity (see also [1, Section 11.6]). This proves (22). For the second assertion, Stieltjes integration by parts gives

$$k^n \int_0^{A/k} e^{-kt} dF(t).$$

After the substitution $u = kt$, (22) shows that this tends to

$$e \int_0^A e^{-u} d\left(\frac{u^n}{n!}\right) = e \left(1 - e^{-A} \sum_{j=0}^{n-1} \frac{A^j}{j!} \right).$$

□

Proposition 6.2 (Sharp global upper bound for the extremal norm). *For every fixed $x \in X$,*

$$\limsup_{m \rightarrow \infty} m^n \lambda_m(x) \leq e_x(X). \quad (24)$$

Proof. Retain the embedding and the metric H from Proposition 3.2. First take $m = rk$ and restrict the normalized coherent state $S_{k,x}$ to X . Choose Ω -normal coordinates centered at x , put $q = \pi|z|^2$ and $\beta = dd^c q$, and use a preferred local frame. The near-diagonal statement in Lemma 5.1 says, for each fixed $A > 0$,

$$|S_{k,x}(z)|_{H^k}^2 = e^{-kq(z)}(1 + o_A(1)) \quad \text{uniformly when } q(z) < A/k.$$

Moreover, $\Omega = \beta + O(|z|)$ in these coordinates. Since $\Omega|_X = r\omega$, Lemma 6.1 gives

$$\begin{aligned} (rk)^n \int_{X_{\text{reg}} \cap \{q < A/k\}} |S_{k,x}|_{h^{rk}}^2 \frac{\omega^n}{n!} \\ \longrightarrow e_x(X) \left(1 - e^{-A} \sum_{j=0}^{n-1} \frac{A^j}{j!} \right). \end{aligned} \quad (25)$$

The off-diagonal estimate and Lemma 5.2 yield

$$\lim_{A \rightarrow \infty} \limsup_{k \rightarrow \infty} (rk)^n \int_{d_\Omega(x,y) \geq c_1 \sqrt{A/k}} |S_{k,x}(y)|_{h^{rk}}^2 \frac{\omega^n}{n!} = 0.$$

Indeed, after splitting into annuli, the right-hand side is bounded by a constant times $\int_{c_2\sqrt{A}}^{\infty} e^{-ct} t^{2n-1} dt$. Letting $A \rightarrow \infty$ in (25) therefore gives

$$(rk)^n \|S_{k,x}|_X\|_{rk}^2 \longrightarrow e_x(X). \quad (26)$$

For a residue class modulo r , fix $a_j \equiv j \pmod{r}$ and choose a fixed section $\tau_j \in H^0(X, L^{a_j})$ with $|\tau_j(x)|_{h^{a_j}} = 1$. For $m = rk + a_j$, use $\tau_j(S_{k,x}|_X)$. On the $k^{-1/2}$ scale its squared norm factor $|\tau_j|^2$ converges uniformly to 1, while it is globally bounded and does not affect the off-diagonal tail. Since $m/(rk) \rightarrow 1$,

$$\limsup_{\substack{m \rightarrow \infty \\ m \equiv j \pmod{r}}} m^n \lambda_m(x) \leq e_x(X).$$

There are only finitely many residue classes, proving (24). $\square$

Combining Propositions 4.4 and 6.2 gives

$$m^n \lambda_m(x) \longrightarrow e_x(X).$$

The extremal identity (2) now proves Theorem 2.4.

7 Examples, other measures, and normalized volume

7.1 The projective quadric cone

Consider

$$X = \{Z_0 Z_2 = Z_1^2\} \subset \mathbb{P}^3, \quad L = \mathcal{O}_{\mathbb{P}^3}(1)|_X,$$

with the restricted Fubini–Study metric, and let $v = [0 : 0 : 0 : 1]$ be the vertex. The germ at v is an A_1 singularity and $e_v(X) = 2$. In the affine chart $Z_3 = 1$, put $r^2 = |z_0|^2 + |z_1|^2 + |z_2|^2$. The section $\sigma_m = Z_3^m$ satisfies

$$|\sigma_m|_{h^m}^2 = (1 + r^2)^{-m}.$$

The circle rotating (Z_0, Z_1, Z_2) and fixing Z_3 preserves the metric and the volume. Hence σ_m is orthogonal to every section vanishing at v , and it is the minimum-norm section with value one at v . A radial calculation on the homogeneous cone gives

$$\int_{\{r^2 < T\}} \frac{\omega^2}{2} = \left(\frac{T}{1+T} \right)^2,$$

and therefore

$$\|\sigma_m\|_m^2 = \int_0^\infty (1+T)^{-m} d\left(\frac{T}{1+T} \right)^2 = \frac{2m!}{(m+2)!}, \quad B_m(v) = \frac{(m+1)(m+2)}{2}.$$

Thus

$$\lim_{m \rightarrow \infty} \frac{B_m(v)}{m^2} = \frac{1}{2} = \frac{1}{e_v(X)}.$$

This example already rules out the statement that the limit should equal the multiplicity itself.

7.2 Why an adapted measure can have a different scale

The A_1 germ admits the quotient parametrization

$$(s, t) \mapsto (s^2, st, t^2).$$

The pullback of a coarse smooth Kähler potential is quadratic in the invariant coordinates and hence behaves like $(|s|^2 + |t|^2)^2$. The coarse Kähler volume carries the corresponding degeneracy and gives the m^{-2} norm scale above. By contrast, an adapted canonical measure pulls back to a smooth positive multiple of Lebesgue measure. Consequently,

$$\int_{\mathbb{C}^2} e^{-cm(|s|^2 + |t|^2)^2} d\lambda_4 \asymp m^{-1}.$$

For this measure the vertex Bergman function has order m , not m^2 . Thus Theorems 2.3 and 2.4 are specifically statements about the matched coarse volume $\omega^n/n!$.

7.3 Comparison with normalized volume

For a klt singularity, Chi Li's normalized volume is

$$\widehat{\text{Vol}}(x, X) = \inf_{v \in \text{Val}_{X,x}} A_X(v)^n \text{Vol}(v).$$

It is generally not the invariant appearing in Theorem 2.4. For example,

$$A_{r-1} = \{xy = z^r\} \simeq \mathbb{C}^2/\mu_r$$

has

$$\text{mult}_0 A_{r-1} = 2, \quad \widehat{\text{Vol}}(0, A_{r-1}) = \frac{4}{r}.$$

The first number is independent of r , while the second follows from the finite-degree formula for normalized volumes [9] applied to the degree- r crepant quotient $\mathbb{C}^2 \rightarrow \mathbb{C}^2/\mu_r$. Therefore the coarse-smooth Bergman coefficient is always $1/2$ in this family, whereas the normalized volume varies with r .

There is nevertheless a different metric setting in which normalized volume naturally appears. For a quotient equipped with its nondegenerate orbifold metric and standard orbifold volume, the Bergman coefficient at the fixed point is $|G|$ when the stabilizer acts trivially on the line-bundle fiber; this is a special case of the orbifold expansion in [3, Theorem 1.4]. Since

$$\widehat{\text{Vol}}(0, \mathbb{C}^n/G) = \frac{n^n}{|G|},$$

for a quasi-étale quotient, by the finite-degree formula [9], this coefficient equals $n^n/\widehat{\text{Vol}}$. More generally, on a noncollapsed Kähler–Einstein Fano limit, define the metric density by

$$\Theta(x) = \lim_{\rho \downarrow 0} \frac{\text{Vol}(B_\rho(x))}{(\pi^n/n!) \rho^{2n}}.$$

In this setting normalized volume is related to $\Theta(x)$ by

$$\widehat{\text{Vol}}(x, X) = n^n \Theta(x).$$

See [5, Theorem 5.5 and Corollary 5.7] for this relation in the noncollapsed Kähler–Einstein Fano limit setting. The tangent-cone Fock model then predicts the coefficient $\Theta(x)^{-1} = n^n/\widehat{\text{Vol}}(x, X)$. This is a different problem: the Kähler–Einstein or orbifold metric is singular from the coarse smooth viewpoint used in this note. Outside the orbifold case, a general theorem on pointwise Bergman localization at such a singular metric tangent cone is not currently supplied by the standard singular-space equidistribution results.

Remark 7.1 (Cartier versus orbifold line bundles). *The residue-class argument in the proof uses that L is an honest line bundle on the coarse space. For an orbifold or merely $\mathbb{Q}$ -Cartier line bundle with nontrivial stabilizer character at x , every section can vanish at x in forbidden congruence classes. The full sequence $m^{-n}B_m(x)$ may then be periodic or may fail to converge.*

8 Concluding remarks

The argument proves more than the order estimate but less than a global singular Tian–Catlin–Zelditch expansion. At each fixed point it identifies the leading coefficient by the algebraic tangent cone. Near a singular stratum, however, there is an $m^{-1/2}$ transition region, and no expansion with coefficients continuous across the singular locus should be expected. Every individual B_m is nevertheless smooth in the ambient complex-space sense. Moreover, the transition region creates no leading order overshoot: the optimal global upper constant satisfies

$$C_m^{\text{opt}} = 1 + O(m^{-1}).$$

Thus its limit is the smooth-locus coefficient 1, even though the normalized Bergman functions do not converge uniformly to their multiplicity-dependent pointwise limit.

The existing literature on Bergman kernels over normal Kähler spaces gives powerful logarithmic and current-level convergence statements [2], but it does not by itself identify the value at a fixed singular point, which is a set of measure zero. For that reason the two complementary extremal estimates above are essential: Lelong–Jensen prevents a section from concentrating more strongly than the multiplicity permits, while ambient coherent states attain the local Gaussian mass asymptotically.

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